

AI Security Lab — Level 05: Rules Collide

Vulnerability Overview

- **Vulnerability Type:** Prompt Conflict / Instruction Hierarchy Manipulation
- **Objective:** Evaluate how the AI system handles conflicting instructions and determine whether user-controlled input can influence or override higher-priority behavioral constraints.

Technical Analysis

AI applications commonly process multiple sources of instructions and contextual information, including system instructions, developer instructions, conversation history, tool outputs, and user input.

When these sources contain conflicting instructions, the model's behavior depends on the application's instruction hierarchy, model behavior, prompt architecture, and implemented security controls.

A security weakness may occur when the application relies on ambiguous natural-language instructions without sufficient structural separation between trusted instructions and untrusted user content.

Security Impact

Potential impacts include:

- Circumvention of intended behavioral restrictions.
- Unauthorized disclosure of protected information.
- Manipulation of model behavior.
- Unexpected tool invocation or downstream actions when the model is connected to external capabilities.

Remediation

- Implement **structured input handling** that clearly separates trusted instructions from untrusted user content.
- Maintain an explicit instruction hierarchy.
- Do not rely solely on natural-language instructions to enforce security boundaries.
- Implement authorization controls independently at the application and API layers.
- Apply tool-level permission checks before allowing the model to perform privileged operations.

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